

SANGJUN (SAMUEL) Ko

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EDUCATION

University of Illinois at Urbana-Champaign	Champaign, Illinois
Bachelor of Science in Statistics (CGPA: 3.83/4.00)	January 2023 - May 2025 (Expected)
• Mathematics & Data Science Minor	
University of Michigan-Shanghai Jiao Tong University Joint Institute	Shanghai, China
Bachelor of Engineering in Mechanical Engineering (CGPA: 3.50/4.00)	September 2019 - December 2020
• First Class Scholarship	

WORK EXPERIENCE

Applied Technologies for Learning in Arts & Sciences	Champaign, Illinois
<i>AI/ML Intern Team Leader</i>	January 2024 - Present
• Lead a team of 5 interns	
• Develop predictive machine learning models with large data sets(100,000+ entries) using Python	
• Utilize advanced techniques: XGBoost, Random Forests, Regression, Neural Networks, BERT, Transformers	
• Collaborate closely with 3 coordinators in weekly meetings to align project goals and track progress	
STAT107 Data Science Discovery	Champaign, Illinois
<i>Course Assistant</i>	August 2023 - Present
• Facilitate learning for over 1000 students by assisting students in grasping fundamental data science concepts by conducting interactive sessions, and providing clarifications on course materials	
• Manage Git for learning platform modifications, benefiting 1000+ students; create, edit, and proofread course materials	
• Collaborate with a team of 100+ staff and 2 professors in weekly meetings	
KM Internacional S.A.	Managua, Nicaragua
<i>Intern</i>	May 2023 - August 2023
• Oversaw comprehensive inventory management and streamlined administrative tasks for a small textile company	
• Lead a team of three workers to achieve a 98% accuracy rate in record-keeping and a 15% reduction in operational costs	

PROJECTS/AWARDS

Graduation Prediction Model	September 2024 - December 2024
• Increased AUC score from 0.65 to 0.82 by reducing noise in the dataset through feature engineering and outlier removal across 50,000+ student records.	
• Built an interactive Streamlit interface with 12 key predictors, enabling real-time graduation predictions for university administrators.	
3rd Place, 2024 COVID Information Commons Student Paper Challenge	September 2024
• Analyzed the socioeconomic and mental health impacts of COVID-19 in Mexico, identifying that 42% of females and 40% of young adults faced job loss	
• Proposed data-driven interventions to support vulnerable groups through geospatial and statistical analysis	
Paraphrase Detection	July 2024
• Achieved 90%+ accuracy in paraphrase detection using BERT, RoBERTa, and SBERT models on the MRPC dataset	
• Developed a Gradio interface for real-time paraphrase detection, facilitating easier evaluation of textual similarities for academic and business use cases	

SKILLS

Languages:	English (Fluent), Spanish (Fluent), Korean (Fluent), French (Basic), Chinese (Basic)
Computer:	Python (Advanced): Pandas, NumPy, Scikit-learn, TensorFlow, NLTK, Matplotlib, Seaborn, BERT, Random Forest, XGBoost, Gradio R (Intermediate): Tidyverse, ggplot2, dplyr, caret Git (Intermediate), SAS (Basic), Tableau (Basic), SQL (Basic), Office (Advanced)
Certificates:	Data Science DISCOVERY, Data Science Certificate, Coursera Business Analytics Certificate

COMMUNITY SERVICE AND EXTRACURRICULAR ACTIVITIES

Consultant in Illinois Business Consulting	September 2024 - Present
• Collaborating with interdisciplinary teams to deliver strategic solutions for real-world business challenges, including market analysis, operational optimization, and growth strategy, for clients ranging from Fortune 100 companies to nonprofits	